Entropy-Weighted Local Concept Matching for Zero-Shot Out-of-Distribution Detection

Anonymous Author(s)
Affiliation
Address
email

Abstract

Reliable out-of-distribution detection is critical for safe machine learning deployment where unknown classes naturally emerge. While vision-language models like CLIP enable promising zero-shot OOD detection, existing methods rely on global image representations corrupted by irrelevant backgrounds, causing suboptimal performance. We propose Entropy-Weighted Local Concept Matching (ELCM), enhancing OOD detection through intelligent local patch aggregation with entropy-based weighting and class-conditional scaling. Our method introduces three innovations: (1) entropy-weighted patch selection focusing on low-confusion regions while suppressing noise, (2) class-conditional scaling amplifying patches with clear preferences, and (3) top-K selection with percentile-based weight stabilization. Extensive experiments demonstrate ELCM achieves superior performance across diverse OOD types, with strong fine-grained recognition results (97.5% AUROC on iNaturalist). Overall, our method attains 91.9% AUROC and 29.8% FPR95, representing a substantial 5.2 percentage point FPR95 reduction versus the strong GL-MCM baseline. This improvement directly translates to enhanced deployment reliability. Comprehensive ablations reveal each component contributes meaningfully, with entropy weighting and class-conditional scaling being particularly crucial.

1 Introduction

Out-of-distribution (OOD) detection identifies test samples from unseen classes (Hendrycks & Gimpel, 2017; Huang & Li, 2021), crucial for safe deployment. Traditional methods require labeled in-distribution data (Lee et al., 2018; Liang et al., 2018; Liu et al., 2020), often using outlier exposure (Hendrycks et al., 2019). Vision-language models like CLIP (Radford et al., 2021) enable zero-shot OOD detection using only class names (Fort et al., 2021; Esmaeilpour et al., 2021).

Zero-shot OOD detection leverages vision-language models to assess whether an image belongs to known classes (Ming et al., 2022; Esmaeilpour et al., 2021). CLIP's joint embedding space enables direct comparison between image features and textual descriptions. Recent methods like Maximum Concept Matching (MCM) (Ming et al., 2022) and Global-Local MCM (GL-MCM) (Miyai et al., 2025) compute similarity scores between global image features and class text embeddings.

However, existing approaches struggle with complex images where global representations are corrupted by irrelevant backgrounds, causing false confidence in OOD samples. Local patch analysis offers solutions but naive aggregation fails as patches vary in informativeness.

We propose Entropy-Weighted Local Concept Matching (ELCM), addressing these challenges through entropy-based patch filtering, class-conditional scaling, and top-K selection with percentile-based weight stabilization. Our core insight is that effective OOD detection requires focusing on discriminative regions while suppressing irrelevant patches.

Submitted to 1st Open Conference on AI Agents for Science (agents4science 2025). Do not distribute.

Our experimental evaluation demonstrates the effectiveness of this approach, achieving substantial improvements over strong baselines with an overall AUROC of 91.9% and FPR95 of 29.8% compared to GL-MCM's 91.3% AUROC and 35.0% FPR95. The 5.2 percentage point reduction in false positive rate directly translates to improved deployment reliability in practical applications.

Our contributions include:

- We propose ELCM, intelligently aggregating local patch features through entropy-based filtering, class-conditional scaling, and percentile-based weight stabilization.
- We introduce an uncertainty-aware framework addressing max pooling limitations through principled patch selection and aggregation.
- We achieve superior performance (91.9% AUROC, 29.8% FPR95), representing 5.2 percentage point FPR95 reduction over GL-MCM, with particularly strong fine-grained recognition results (97.5% AUROC on iNaturalist).
- We provide extensive ablations demonstrating component synergy and revealing that entropy filtering and class-conditional scaling drive the primary improvements.

2 Related Work

Traditional OOD detection methods (Hendrycks & Gimpel, 2017; Liang et al., 2018; Liu et al., 2020) require task-specific training, limiting zero-shot applicability. Vision-language models like CLIP (Radford et al., 2021) enable zero-shot OOD detection (Fort et al., 2021; Ming et al., 2022). Early methods used negative prompts (Fort et al., 2021; Esmaeilpour et al., 2021) but faced scalability issues. Maximum Concept Matching (MCM) (Ming et al., 2022) improved through global image-text similarities, with advances including CLIPN (Wang et al., 2023) and NPOS (Tao et al., 2023).

Citation: NPOS is not related to MCM.

GL-MCM (Miyai et al., 2025) incorporates local patch features using max pooling: $S_{\text{local}} = \max_{i,j} \frac{\exp(\text{sim}(\mathbf{l}_i, \mathbf{t}_j)/\tau)}{\sum_{k=1}^{K} \exp(\text{sim}(\mathbf{l}_i, \mathbf{t}_k)/\tau)}$. This suffers from spurious high-confidence patches and lacks mechanisms to suppress confused regions.

This equation is correct, but there is no detailed explanation.

Entropy provides reliable uncertainty measurement (Lakshminarayanan et al., 2017; Ren et al., 2019), revealing informative versus noisy regions in vision transformers (Dosovitskiy et al., 2021). However, zero-shot OOD detection has not systematically leveraged entropy for patch selection.

Our Contribution. We propose ELCM with principled local aggregation through entropy-based filtering, class-conditional scaling, and weight stabilization, ensuring only informative patches contribute to decisions.

3 Method

3.1 Overview

We present Entropy-Weighted Local Concept Matching (ELCM), enhancing vision-language OOD detection through intelligent local feature aggregation. Building upon GL-MCM (Miyai et al., 2025), our method improves how local patch features are selected, weighted, and aggregated by focusing on discriminative, confident regions while suppressing noise from irrelevant patches.

3.2 Global-Local Maximum Concept Matching (GL-MCM)

Our work extends the GL-MCM baseline (Miyai et al., 2025), which combines global and local CLIP features for OOD detection. Given an input image, GL-MCM extracts both global features $\mathbf{g} \in \mathbb{R}^d$ from the CLS token and local features $\mathbf{L} = \{\mathbf{l}_i\}_{i=1}^{N} \in \mathbb{R}^{N \times d}$ from patch tokens of the Vision Transformer backbone (Dosovitskiy et al., 2021), where N is the number of patches and d is the feature dimension. For a set of K in-distribution class names, text features $\mathbf{T} = \{\mathbf{t}_j\}_{j=1}^{K} \in \mathbb{R}^{K \times d}$ are extracted using CLIP's text encoder (Radford et al., 2021).

The global score is computed as:

$$S_{\text{global}} = \max_j \frac{\exp(\text{sim}(\mathbf{g}, \mathbf{t}_j)/\tau)}{\sum_{k=1}^{K} \exp(\text{sim}(\mathbf{g}, \mathbf{t}_k)/\tau)} \tag{1}$$

The local score uses simple max pooling:

$$S_{\text{local}} = \max_{i,j} \frac{\exp(\text{sim}(\mathbf{l}_i, \mathbf{t}_j)/\tau)}{\sum_{k=1}^{K} \exp(\text{sim}(\mathbf{l}_i, \mathbf{t}_k)/\tau)} \tag{2}$$

The final GL-MCM score combines both components:

$$S_{\text{GL-MCM}} = S_{\text{global}} + \lambda S_{\text{local}} \tag{3}$$

where τ is the temperature parameter and λ controls the relative importance of local features. While GL-MCM shows improvements over purely global methods, its simple max pooling aggregation can be dominated by spurious high-confidence patches and fails to exploit the rich structure in local feature distributions.

3.3 Entropy-Weighted Local Concept Matching (ELCM)

As an overall impression of the proposed method, there is little convincing explanation for why each step is necessary or why the chosen parameter settings are sufficient. It feels more like a solution produced through the agent's trial-and-error process than a well-justified design.

We propose ELCM to address the limitations of naive local feature aggregation through three key innovations: entropy-based patch filtering, class-conditional scaling, and top-K selection with percentile-based weight stabilization.

Entropy-Based Patch Filtering. Instead of treating all patches equally, we use entropy to identify and suppress highly confused regions. For each patch i, we compute the probability distribution over classes:

$$p_{i,j} = \frac{\exp(\text{sim}(\mathbf{l}_i, \mathbf{t}_j)/\tau)}{\sum_{k=1}^{K} \exp(\text{sim}(\mathbf{l}_i, \mathbf{t}_k)/\tau)} \tag{4}$$

The entropy of patch i is:

$$H_i = -\sum_{j=1}^{K} p_{i,j} \log p_{i,j} \tag{5}$$

High entropy indicates confusion or ambiguity, suggesting the patch contains uninformative content. We filter patches using an entropy threshold H_{thresh}, automatically computed as the 75th percentile of patch entropies to remove the most confused regions.

Class-Conditional Scaling. To further enhance discrimination, we introduce class-conditional scaling that amplifies patches with clear class preferences. We compute a discrimination ratio based on the top-K_c class probabilities:

$$r_i = \frac{\max_j p_{i,j}}{\frac{1}{K_c} \sum_{j \in \text{top-}K_c} p_{i,j}} \tag{6}$$

The class-conditional factor is:

$$\gamma_i = r_i^{\beta} \tag{7}$$

where β controls the strength of class-conditional scaling. This factor amplifies patches that strongly prefer a single class while dampening those with uniform distributions across multiple classes.

Top-K Selection and Percentile-Based Weight Stabilization. After entropy filtering, we select the top-K patches based on class-conditional scaled confidence:

$$c_i = \max_j p_{i,j} \cdot \gamma_i \tag{8}$$

For the selected patches, we apply percentile-based weight stabilization instead of naive exponential entropy weighting. We compute the 25th and 75th percentiles of entropies among selected patches, then assign weights as:

$$w_i = \begin{cases} 1.0 & \text{if } H_i \leq H_{25} \\ 1.0 - \frac{H_i - H_{25}}{H_{75} - H_{25}} \cdot (1.0 - \gamma_{\min}) & \text{if } H_{25} < H_i < H_{75} \\ \gamma_{\min} & \text{if } H_i \geq H_{75} \end{cases} \tag{9}$$